

Exercise 8.1 Suppose one can send M distinct messages through n -copies of a memoryless channel W with average error at most η . Use Fano's inequality to show that

$$\log M \leq \frac{nI(W) + 1}{1 - \eta}.$$

Solution: Let M and $\hat{M}$ be random variables denoting the input and the output of the above process, and let X_k and Y_k be random variables denoting the channel input and output at k -th channel use. By assumption, we know

- $H(M) = \log M$
- $M - (X_1, \dots, X_n) - (Y_1, \dots, Y_n) - \hat{M}$ forms a Markov chain
- $P\{M \neq \hat{M}\} \leq \eta$

By Fano's inequality, we have

$$I(M : \hat{M}) = H(M) - H(M|\hat{M}) \geq H(M) - (1 + P\{M \neq \hat{M}\} \cdot \log M) \geq (1 - \eta) \cdot \log M - 1.$$

Due to the Markov chain property, we know $I(M : \hat{M}) \leq I(X_1 \dots X_n : Y_1 \dots Y_n)$, thus

$$(1 - \eta) \cdot \log M - 1 \leq I(X_1 \dots X_n : Y_1 \dots Y_n).$$

Thus,

$$\log M \leq \frac{I(X_1 \dots X_n : Y_1 \dots Y_n) + 1}{1 - \eta} \leq \frac{I(W^n) + 1}{1 - \eta} = \frac{nI(W) + 1}{1 - \eta},$$

where we used the additivity of the channel mutual information for DMCs in the last step.

Exercise 8.2 In this exercise, we will derive a sample complexity for distribution learning with respect to ℓ_2 -norm: instead of requiring $\delta_{\text{tvd}}(P, \hat{P}_{X^n}) \leq \epsilon$, we would like to guarantee $\|P - \hat{P}_{X^n}\|_2 \leq \epsilon$. We are also going to explicitly include the scaling in the probability of error, η , in our bound.¹

a.) Show that the empirical estimator achieves $P[\|P - \hat{P}_{X^n}\|_2 \leq \epsilon] \geq 1 - \eta$ with

$$n = O\left(\frac{d}{\epsilon^2} \log\left(\frac{d}{\eta}\right)\right)$$

Hint: Note that the approach from the lecture would result in $O(d/\epsilon^2\eta)$. Instead, you may ensure that none of the events $L(x) := \mathbf{1}\{|P(x) - \hat{P}_{X^n}(x)| \geq \epsilon/\sqrt{d}\}$ take place and then apply a Hoeffding bound, i.e. for independent random variables $X_1 \dots X_n$ satisfying $X_i \in [a_i, b_i]$ and $S_n = \sum_{i=1}^n X_i$

$$P[S_n - \mathbb{E}[S_n] \geq t] \leq e^{-\frac{t^2}{\sum_{i=1}^n (b_i - a_i)^2}}.$$

b.) It turns out that the above bound on the sample complexity is not tight. In fact, it can be shown that learning in ℓ_2 is possible using only $n = O(\log(1/\eta)/\epsilon^2)$ samples - without dependence on the dimension of the distribution, a stark contrast to learning in δ_{tvd} ! Assume such an algorithm is given: argue that the scaling in ϵ , $O(1/\epsilon^2)$, of this algorithm has to be tight. Assume that there exists a more efficient learner using only $O(1/\epsilon^\alpha)$ samples, for $\alpha < 2$ and find a contradiction.

Hint: In the lecture you have seen that for $u, v \in \mathbb{R}^d$, $\|u \cdot v\|_1 \leq \|u\|_2 \|v\|_2$. How do you have to choose u and v to obtain a relation between $\delta_{\text{tvd}}(P, \hat{P}_{X^n})$ and $\|P - \hat{P}_{X^n}\|_2$?

¹If you are interested in more details, see [arXiv:2002.11457](https://arxiv.org/abs/2002.11457). The same paper also proves the tight sample complexity for learning in variation distance including η , which is in $\Theta([d + \log(1/\eta)]/\epsilon^2)$.

Solution:

a.) We first rewrite

$$\begin{aligned} P[\|P - \hat{P}_{X^n}\|_2 \geq \epsilon] &= P\left[\sqrt{\sum_x (P(x) - \hat{P}_{X^n}(x))^2} \geq \epsilon\right] = P\left[\sum_x (P(x) - \hat{P}_{X^n}(x))^2 \geq \epsilon^2\right] \\ &\leq P\left[\bigvee_x \left((P(x) - \hat{P}_{X^n}(x))^2 \geq \frac{\epsilon^2}{d}\right)\right] = P\left[\bigvee_x \left(|P(x) - \hat{P}_{X^n}(x)| \geq \frac{\epsilon}{\sqrt{d}}\right)\right]. \end{aligned}$$

We now apply a union bound. If we can ensure that the resulting sum is bounded by η , then this implies the required guarantee on $P[\|P(x) - \hat{P}_{X^n}(x)\|_2]$.

$$P\left[\sum_x |P(x) - \hat{P}_{X^n}(x)| \geq \epsilon\right] \leq \sum_x P\left[|P(x) - \hat{P}_{X^n}(x)| \geq \epsilon/\sqrt{d}\right] \stackrel{!}{\leq} \eta.$$

To apply the Hoeffding bound, we note that $\hat{P}_{X^n}(x) = \frac{1}{n} \sum_{k=1}^n X_k(x)$, where $X_k(x)$ is an indicator variable denoting whether we observe label x in the k th sample. Note that for each term $a_k := 0 \leq X_k(x)/n \leq 1/n =: b_k$, such that

$$P[|P(x) - \hat{P}_{X^n}(x)| \geq \epsilon/d] \leq 2 \exp\left(-\frac{\epsilon^2/d}{\sum_k (a_k - b_k)^2}\right) = 2 \exp\left(-\frac{\epsilon^2 n^2}{dn}\right).$$

Since this is independent of x we can bound

$$P\left[\sum_x |P(x) - \hat{P}_{X^n}(x)| \geq \epsilon\right] \leq 2d \exp(-n\epsilon^2/d).$$

Choosing

$$n = \frac{d}{\epsilon^2} \log\left(\frac{2d}{\eta}\right)$$

shows the desired bound. Note that depending on the relationship between d and η , this sample complexity could be worse than the one obtained from the approach presented in the lecture.

b.) We first relate $\delta_{\text{tvd}}(P, \hat{P}_{X^n})$ and $\|P - \hat{P}_{X^n}\|_2$. For this we define $u(x) := P(x) - \hat{P}_{X^n}(x)$ and $v(x) = 1$ for all x . Then

$$\|u \cdot v\|_1 \leq \|u\|_2 \|v\|_2 \implies \|P - \hat{P}_{X^n}\|_1 \leq \sqrt{d} \|P - \hat{P}_{X^n}\|_2.$$

Assume we had a learner L which is more efficient than $O(1/\epsilon^2)$ samples, $O(1/\epsilon^\alpha)$, where $\alpha < 2$ (recall that we neglect the scaling in η here). We first learn the distribution to precision $\tilde{\epsilon} := \epsilon/\sqrt{d}$ in l_2 distance using L . We can then use the above relation between the different norms,

$$\delta_{\text{tvd}}(P, \hat{P}_{X^n}) \leq \sqrt{d} \|P - \hat{P}_{X^n}\|_2,$$

which implies that $\hat{P}_{X^n}$ also satisfies $\delta_{\text{tvd}}(P, \hat{P}_{X^n}) \leq \epsilon$. However, we would have achieved this using $O(1/\tilde{\epsilon}^\alpha) < O(d/\epsilon^2)$ samples, a violation of the lower bound derived in the lecture. We thus conclude that such an algorithm cannot exist.

Exercise 8.3 Consider the same setup as in Theorem 8.2 but replacing $\mathcal{P}$ by the class of environments with k arms that allows for rewards with Gaussian bounded variance, i.e if $\nu \in \mathcal{P}$ then

$$\nu = \{\mathcal{N}(\cdot; \mu_a, \sigma_a^2) : a \in [k], \mu_a \in [0, 1], \sigma_{\min}^2 \leq \sigma_a^2 \leq \sigma_{\max}^2\},$$

where $0 \leq \sigma_{\min} \leq \sigma_{\max}$.

a.) Show that

$$R_n^*(\mathcal{P}) \geq \frac{\sigma_{\max}}{27} \sqrt{kn}.$$

b.) Let $\sigma_{\max}^2 = n'$ for some fixed $n' \in \mathbb{N}$. What is the behaviour of $R_n^*(\mathcal{P})$ for $n \leq n'$? Does it get worse or better the scaling of the minimax regret on the time horizon n ? What happens to $R_n^*(\mathcal{P})$ if we allow $\sigma_{\max}^2 \approx 0$?

Hint: You can follow the thread in the lecture notes, without assuming the variance to be 1.

Discuss your ideas at a high-level understanding the role that play the variance when learning from distributions.

Solution:

a.) We fix a policy π and following Section 8.2.4 we consider the environments

$$\nu_\mu = \{\mathcal{N}(\cdot; \mu_1, \sigma_{\max}^2), \mathcal{N}(\cdot; \mu_2, \sigma_{\max}^2), \dots, \mathcal{N}(\cdot; \mu_k, \sigma_{\max}^2)\},$$

where $\mu \in [0, 1]^k$. Then following the lecture notes we define analogously the environments. First we fix the environment ν_μ with

$$\mu = (\Delta, 0, \dots, 0),$$

for some $\Delta \in (0, 1]$. Then letting π run on ν_μ we define

$$i_* := \operatorname{argmin}_{i \in [k]} \{\mathbb{E}[T_n(i)]\},$$

and the environment $\nu_{\mu'}$ with

$$\mu' = (\Delta, 0, \dots, 0, \underbrace{2\Delta}_{\text{at position } i_*}, 0, \dots, 0).$$

Then the exercise follows as in the lecture notes until in Equation 8.72 where we have to evaluate the divergence of the probability distributions p, p' induced by the environments $\nu_\mu, \nu_{\mu'}$ respectively. In this case we get

$$D(p \| p') = \mathbb{E}[T_{i_*}(n)] D(\mathcal{N}(\cdot; 0, \sigma_{\max}^2) \| \mathcal{N}(\cdot; 2\Delta, \sigma_{\max}^2)).$$

Using that for gaussian distributions with same variance the relative entropy can be computed as

$$D(\mathcal{N}(\cdot; \mu_1, \sigma^2) \| \mathcal{N}(\cdot; \mu_2, \sigma^2)) = \frac{(\mu_1 - \mu_2)^2}{2\sigma^2},$$

we have

$$D(p \| p') \leq \frac{2\Delta^2 n}{\sigma_{\max}^2 (k-1)},$$

where we used that $\mathbb{E}[T_{i_*}(n)] \leq n/(k-1)$ due to the pigeonhole principle (Equation 8.45 from lecture notes). Then we have

$$R_n(\pi, \nu_\mu) + R_n(\pi, \nu_{\mu'}) \geq \frac{n\Delta}{4} \exp\left(-\frac{2\Delta^2 n}{\sigma_{\max}^2 (k-1)}\right).$$

The exercise follows choosing

$$\Delta = \sigma_{\max} \sqrt{\frac{k-1}{4n}}.$$

b.) In the first scenario we get

$$R_n^*(\mathcal{P}) \geq \frac{\sqrt{k}}{27}n$$

since $\sigma_{\max} \geq \sqrt{n'} \geq \sqrt{n}$. We get a linear dependence on n for $n \leq n'$ which is worse than the $\sqrt{n}$. This makes sense since now the variance of the distributions is higher and this makes harder to learn the means for each arm of the bandit since the samples will be "less concentrated" around the true mean.

In the other case we have $R_n^*(\mathcal{P}) \approx 0$. In this case the problem becomes easier since now the samples of each arm will be "very concentrated" around the true mean and this will allow to find faster the best arms.

Exercise 8.4 In this exercise we are going to derive a minimax regret lower bound for a two-armed bandit with Bernoulli rewards. Let's denote $\text{Bern}(\cdot; \mu)$ the pdf of a Bernoulli distribution with support in $\{0, 1\}$ and mean $\mu \in [0, 1]$ i.e

$$\text{Bern}(x; \mu) = \begin{cases} \mu & \text{if } x = 1 \\ 1 - \mu & \text{if } x = 0 \\ 0 & \text{else} \end{cases}$$

The construction will be very similar to the one from the lecture notes and it involves the computation of a relative entropy between Bernoulli distributions. In order to ease this step we are going to use a "simpler" quantity that is the χ^2 -distance which is defined as

$$\chi^2(p, q) := \int_{\mathcal{X}} \frac{p^2(x)}{q(x)} dx - 1,$$

for two pdfs p and q of the same random variable X taking values on $\mathcal{X}$.

a.) Show the following inequality between the relative entropy and the χ^2 -distance

$$\frac{1}{\log(e)} D(p||q) \leq \chi^2(p, q).$$

Hint: Use the inequality $\ln(1+x) \leq x$ for $x > -1$.

b.) Show that the χ^2 -distance for two Bernoulli's with means μ, μ' is given by

$$\chi^2(\text{Bern}(\cdot; \mu), \text{Bern}(\cdot; \mu')) = \frac{(\mu - \mu')^2}{\mu'(1 - \mu')}.$$

c.) Consider now $\mu \in [0, 1]^2$ and the two-armed Bernoulli environment ν_μ given by

$$\nu_\mu = \{\text{Bern}(\cdot; \mu_a) : a \in [2]\} \quad \text{where } \mu = \left(\frac{1}{2} + \frac{\Delta}{4}, \frac{1}{2}\right)$$

for some constant $\Delta \in (0, 1]$ to be defined later. Analogously consider $\nu_{\mu'}$ with

$$\mu' = \left(\frac{1}{2} + \frac{\Delta}{4}, \frac{1}{2} + \frac{\Delta}{2}\right).$$

Given a policy π prove that

$$R_n(\pi, \nu_\mu) + R_n(\pi, \nu_{\mu'}) \geq \frac{n\Delta}{16} \exp(-D(p||p')),$$

where p, p' are the probability distributions of rewards and actions induced by the policy π on $\nu_\mu, \nu_{\mu'}$ respectively.

d.) Use a), b) and the trivial bound $\mathbb{E}[T_a(n)] \leq n$ to show

$$D(p||p') \leq \frac{n\Delta^2}{2(1-\Delta^2)}.$$

e.) Combine all the above results and show that for the class of environments with 2 arms and Bernoulli rewards we can bound the minimax regret as

$$R_n^*(\mathcal{P}) \geq \frac{1}{864}\sqrt{n+1}$$

Hint: Solve $n = \frac{1-\Delta^2}{\Delta^2}$ and use $n \geq \frac{n+1}{2}$ for $n \geq 1$.

Solution:

a.) From the definitions and the inequality $\ln(1+x) \leq x$ together with $\log(x)/\log(e) = \ln(x)$ we have

$$\begin{aligned} \frac{1}{\log(e)} D(p||q) &= \int_{\mathcal{X}} p(x) \ln \left(\frac{p(x)}{q(x)} \right) dx \\ &= \int_{\mathcal{X}} p(x) \ln \left(1 + \frac{p(x) - q(x)}{q(x)} \right) dx \\ &\leq \int_{\mathcal{X}} p(x) \frac{p(x) - q(x)}{q(x)} dx \\ &= \int_{\mathcal{X}} \frac{p^2(x)}{q(x)} dx - \int_{\mathcal{X}} p(x) dx \\ &= \int_{\mathcal{X}} \frac{p^2(x)}{q(x)} dx - 1. \end{aligned}$$

b.) From the definitions we have

$$\begin{aligned} \chi^2(\text{Bern}(\cdot; \mu), \text{Bern}(\cdot; \mu')) &= \frac{\mu^2}{\mu'} + \frac{(1-\mu)^2}{1-\mu'} - 1 \\ &= \frac{\mu^2(1-\mu') + (1-\mu)^2\mu' - \mu'(1-\mu')}{\mu'(1-\mu')} \\ &= \frac{\mu^2 - 2\mu\mu' + \mu'^2}{\mu'(1-\mu')} \\ &= \frac{(\mu - \mu')^2}{\mu'(1-\mu')}. \end{aligned}$$

c.) Note that since the environments $\nu_\mu, \nu_{\mu'}$ only have 2 arms the expressions of the regret are simply

$$\begin{aligned} R_n(\pi, \nu_\mu) &= \frac{\Delta}{4} \mathbb{E}[T_2(n)] = \frac{\Delta}{4} (n - \mathbb{E}[T_1(n)]), \\ R_n(\pi, \nu_{\mu'}) &= \frac{\Delta}{2} \mathbb{E}'[T_1(n)] \geq \frac{\Delta}{4} \mathbb{E}'[T_1(n)], \end{aligned}$$

where $\mathbb{E}, \mathbb{E}'$ are taken over p and p' respectively. Now using Markov's inequality we have

$$\begin{aligned} \mathbb{E}[n - T_1(n)] &\geq \frac{n}{2} p \left[n - T_1(n) < \frac{n}{2} \right], \\ \mathbb{E}'[T_1(n)] &\geq \frac{n}{2} p' \left[T_1(n) \geq \frac{n}{2} \right]. \end{aligned}$$

Combining the above expressions lead to

$$R_n(\pi, \nu_\mu) + R_n(\pi, \nu_{\mu'}) \geq \frac{n\Delta}{8} \left(p \left[T_1(n) < \frac{n}{2} \right] + p' \left[T_1(n) \geq \frac{n}{2} \right] \right).$$

The result follows using the Bretagnolle-Huber inequality (Lemma 8.56).

d.) From the divergence decomposition lemma (Lemma 8.5) we have

$$D(p\|p') = \mathbb{E}[T_2(n)] D\left(\text{Bern}(\cdot; \frac{1}{2})\|\text{Bern}(\cdot; \frac{1}{2} + \frac{\Delta}{2})\right).$$

Using a) and $\mathbb{E}[T_2(n)] \leq n$

$$D(p\|p') \leq n\chi^2\left(\text{Bern}(\cdot; \frac{1}{2}), \text{Bern}(\cdot; \frac{1}{2} + \frac{\Delta}{2})\right).$$

The result follows using b) and the explicit computation

$$\begin{aligned}\chi^2\left(\text{Bern}(\cdot; \frac{1}{2}), \text{Bern}(\cdot; \frac{1}{2} + \frac{\Delta}{2})\right) &= \frac{(\frac{1}{2} - (\frac{1}{2} + \frac{\Delta}{2}))^2}{(\frac{1}{2} + \frac{\Delta}{2})(\frac{1}{2} - \frac{\Delta}{2})} \\ &= \frac{\Delta^2}{2(1 - \Delta^2)}.\end{aligned}$$

e.) Combining c) and d) with the choice $\Delta = \frac{1}{\sqrt{1+n}}$ we have

$$R_n(\pi, \nu_\mu) + R_n(\pi, \nu_{\mu'}) \geq \frac{n}{16\sqrt{n+1}} \exp\left(-\frac{1}{2}\right).$$

The result follows using $\exp(-1/2) \geq 2/27$ and $n \geq \frac{n+1}{2}$.

Exercise 8.5 *In this exercise we will see a slightly different technique to derive lower bounds for problems in learning theory, which will allow us to find lower bounds for a yes/no question regarding distributions: instead of learning all parameters of a distribution, one might often only be interested in testing for a specific property. For example, consider the problem of uniformity testing: given samples X^n from a distribution, decide whether the underlying distribution is uniform or ϵ far from being uniform.*

We start with the following lemma, which formalizes the intuitive notion that if a variable A carries little information about a fair random bit X , then any attempt to guess X based on observing A can only be marginally better than random.

Lemma. Let X be a fair random bit, A a random variable correlated with X . Then there exist small positive constants c_1, c_2 independent of A as follows: if $I(X : A) \leq c_1$, then for any function f trying to recover X from A , we have $P[f(A) = X] \leq 1/2 + c_2$.

a.) *Prove the lemma. You do not need to determine exact values for c_1, c_2 , just argue that they are strictly larger than zero independent of A . Use that the binary entropy $h_2(p)$ has a single maximum at $p = 1/2$.*

Hint: *It might be more intuitive to show the contrapositive: if there exists f such that $P[f(A) = X] > 1/2 + c_2$, then $I(A : X) > c_1$. Recall that $I(X : Y) := H(X) - H(X|Y)$, and start from writing out the definition of $H(X|f(A))$. Finally use the data processing inequality, $I(X : f(Y)) \leq I(X : Y)$.*

b.) *How can we in principle use this lemma to derive lower bounds for a yes/no question about a distribution?*

Hint: *If A^n is a set of n samples, interpret $I(X : A^n)$ as a function increasing in n , and note that we would like to solve the problem reliably, that is with probability close to 1.*

c.) *For X again a fair random bit and A an arbitrary random variable taking values in an alphabet S_A , show that*

$$I(X : A) \leq \frac{\log(e)}{2} \sum_{a \in S_A} \frac{(P[A = a|X = 0] - P[A = a|X = 1])^2}{P[A = a|X = 0] + P[A = a|X = 1]}.$$

Hint: *Use $\ln(x) \leq x - 1$ (or $\frac{1}{\log(e)} D(P\|Q) \leq \chi^2(P\|Q)$) and the fact that X is fair.*

- d.) Using the above lemma, show a lower bound of $\Omega(1/\epsilon^2)$ for testing whether a coin is fair ($P[\text{Heads}] = 0.5$) or biased by at least ϵ ($|P[\text{Heads}] - 0.5| \geq \epsilon$). We are only interested in the scaling on ϵ .

Hint: Depending on the input, the output distribution of your channel should correspond to either a fair or a biased coin. When applying the result from c.), use that for $x, y > 0$, $(x - y)^2/(x + y) < (x - y)^2/x = x - 2y + y^2/x$, and simplify further using the binomial formula.

Solution:

- a.) We will show the contrapositive and also use that $h(x) < h(1/2)$ for $x \in [0, 1] \setminus \{1/2\}$. First we rewrite $H(X|f(A))$:

$$\begin{aligned} H(X|f(A)) &= - \sum_{x,y \in \{0,1\}} P[X = x, f(A) = y] \log \left(\frac{P[X = x, f(A) = y]}{P[X = x]} \right) \\ &= - \sum_{x \in \{0,1\}} P[X = f(A)|X = x] P[X = x] \log (P[X = f(A)|X = x]) \\ &\quad - \sum_{x \in \{0,1\}} P[X \neq f(A)|X = x] P[X = x] \log (P[X \neq f(A)|X = x]) \end{aligned}$$

Now denote $p_x := P[X = f(A)|X = x]$ (then $1 - p_x = P[X \neq f(A)|X = x]$). Using $P[X = x] = 1/2$ independent of x , we can now write

$$H(X|f(A)) = -\frac{1}{2} \sum_{x \in \{0,1\}} p_x \log(p_x) + (1 - p_x) \log(1 - p_x) = \frac{1}{2} (h_2(p_0) + h_2(p_1)).$$

From the assumption $P[f(A) = X] > 1/2 + c_2$, we find

$$1/2 + c_2 < P[f(A) = X|X = 0]P[X = 0] + P[f(A) = X|X = 1]P[X = 1] = \frac{p_0 + p_1}{2}.$$

This means that $p_0 \neq 1/2 \vee p_1 \neq 1/2$. Since the binary entropy is maximal for $p = 1/2$, it follows that

$$H(X|f(A)) = \frac{1}{2} (h_2(p_0) + h_2(p_1)) < h_2(1/2) = H(X).$$

Finally, we let $c_1 := H(X) - H(X|f(A)) = I(X : f(A))$, and note that $I(X : f(A)) \leq I(X : A)$ from the data processing inequality.

- b.) The idea is to initially select one of two messages based on a fair random bit (for simplicity, call the messages ‘0’ and ‘1’). We then send this message repeatedly through the channel. Depending on which message was selected, the output of the channel will either follow a specific distribution P satisfying the property we want to test for, or a distribution Q which doesn’t satisfy the property.

$$W[y|x] = \begin{cases} P(y) & \text{if } x = 0, \\ Q(y) & \text{if } x = 1. \end{cases}$$

A valid decoder aiming to retrieve X would then be a learning algorithm which can solve the general decision problem.

It is clear that the mutual information between X and A will increase the more samples we include. Using the lemma, we can increase n until we hit the threshold for a certain n^* , $I(X : A^{n^*}) = c_1$. This means that n^* samples allow us to make a correct guess with probability at most $1/2 + c_2$. However, we would like to make a correct guess with high probability $\gg 1/2 + c_2$: to achieve this, we will need even more samples, and hence n^* is

a lower bound to the sample complexity of our problem.

Note that in general, one would need to modify the above procedure, as we currently only have the possibility to incorporate two different distributions. Distinguishing between them is often not as hard as testing for a property in arbitrary distributions.

- c.) We write out the definition of $I(X : A)$ and then bound the logarithm using $\log(x)/\log(e) = \ln(x) \leq x - 1$. We will also use that $P[X = 0] = P[X = 1] = 1/2$.

$$\begin{aligned}
I(X : A) &= \sum_{\substack{a \in S_A \\ x \in \{0,1\}}} P[A = a, X = x] \log \left(\frac{P[A = a, X = x]}{P[A = a]P[X = x]} \right) \\
&= \sum_{\substack{a \in S_A \\ x \in \{0,1\}}} P[A = a|X = x]P[X = x] \log \left(\frac{P[A = a|X = x]}{P[A = a]} \right) \\
&\leq \log(2) \sum_{\substack{a \in S_A \\ x \in \{0,1\}}} \frac{P[A = a|X = x]^2 - P[A = a|X = x]P[A = a]}{2P[A = a]} \\
&= \log(2) \sum_{\substack{a \in S_A \\ x \in \{0,1\}}} \frac{2P[A = a|X = x]^2 - P[A = a|X = x](P[A = a|X = 0] + P[A = a|X = 1])}{2(P[A = a|X = 0] + P[A = a|X = 1])} \\
&= \frac{\log(2)}{2} \sum_{a \in S_A} \frac{(P[A = a|X = 0] - P[A = a|X = 1])^2}{P[A = a|X = 0] + P[A = a|X = 1]}.
\end{aligned}$$

- d.) The channel will be defined for two inputs, which we will label ‘Fair’ and ‘Biased’. An initial fair random bit will decide which message we send. The channel also has two outputs - they could be seen as bits, but we label them H and T , for ‘Heads’ and ‘Tails’, in order to make the analogy to the coin toss evident. Then we set

$$W[H|\text{Fair}] = 1/2, \quad W[T|\text{Fair}] = 1/2, \quad W[H|\text{Biased}] = 1/2 + \epsilon, \quad W[T|\text{Biased}] = 1/2 - \epsilon.$$

We will send N messages through the channel (which we can see as performing N coin tosses with the unknown coin). We denote by K_H the number of heads we observe, with $K_H \in \{0, 1, \dots, N\}$. Note that K characterizes all information we can get from the outputs. For readability we set $p := 1/2$ in the next step. The last equality follows from the binomial formula.

$$\begin{aligned}
I(X : K_H) &\leq \sum_{k=0}^N \frac{(P[K_H = k|X = \text{Fair}] - P[K_H = k|X = \text{Biased}])^2}{P[K_H = k|X = \text{Fair}]} \\
&= \sum_{k=0}^N \binom{N}{k} \frac{(p^k(1-p)^{N-k} - (p+\epsilon)^k(1-p-\epsilon)^{N-k})^2}{p^k(1-p)^{N-k}} \\
&= \sum_{k=0}^N \binom{N}{k} p^k(1-p)^{N-k} - 2 \sum_{k=0}^N \binom{N}{k} (p+\epsilon)^k(1-p-\epsilon)^{N-k} \\
&\quad + \sum_{k=0}^N \binom{N}{k} \left(\frac{(p+\epsilon)^2}{p} \right)^k \left(\frac{(1-p-\epsilon)^2}{1-p} \right)^{N-k} \\
&= 1 - 2 + \left(\frac{(p+\epsilon)^2}{p} + \frac{(1-p-\epsilon)^2}{1-p} \right)^N
\end{aligned}$$

Now we use that $p = 1/2$, which simplifies everything considerably:

$$\begin{aligned} I(X : K_H) &\leq -1 + \left(\frac{(1/4)^2 + 2(1/2)\epsilon + \epsilon^2}{(1/2)} + \frac{(1/2)^2 - 2(1/2)\epsilon + \epsilon^2}{(1/2)} \right)^N \\ &= -1 + (1 + 4\epsilon^2)^N \end{aligned}$$

The mutual information has to be bounded by the small constant c_1 , which we can ensure by requiring

$$I(X : K_H) \leq -1 + (1 + 4\epsilon^2)^N \stackrel{!}{\leq} c_1 \implies N \log(1 + 4\epsilon^2) \stackrel{!}{\leq} \log(c_1 + 1).$$

Using $\log(y) \leq y - 1$ yields that

$$N^* = \frac{\log(c_1 + 1)}{4\epsilon^2} = \Omega\left(\frac{1}{\epsilon^2}\right)$$

still satisfies the inequality, and is hence a valid lower bound. For this specific problem, the difficulty of answering the yes/no question coincides with learning the full distribution (note that a coin is described by a single parameter). Generalized to d -dimensional distributions, the problem studied here is known as uniformity testing, and can be solved more efficiently (with sublinear scaling in d) than learning the full distribution.